NileCAM81_CUOAGX

e-multicam Application User Manual

Version 1.7

e-con Systems

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Contents

Introduction to NileCAM81_CUOAGX

NileCAM81_CUOAGX is family of NileCAM cameras from e-con Systems with high-speed unified serial interface that carries video data, control data and power in a single coaxial cable. It uses serializer-deserializer (SerDes) technology that allows single coaxial cable interface for the camera. With coaxial cable lengths of up to 15m, NileCAM81_CUOAGX offers greater flexibility for the system architects to place the cameras in mechanically challenging designs of new generation embedded, automotive and autonomous driving applications. The flexibility of coaxial cable and the ability to carry video data, control data and power in a single cable makes NileCAM81_CUOAGX fast (low latency), flexible and yet reliable.

NileCAM81_CUOAGX uses Maxim's Gigabit Multimedia Serial Link (GMSL2) interface technology for the serial interface to the camera. This serial interface uses coaxial cable that carries high-speed video data from the camera, bidirectional control data between the camera and host controller, and power supply for the camera. The power to the camera is supplied from the host processor through this coaxial cable. NileCAM81_CUOAGX embeds the Maxim's serializer which serializes the high-speed image data and transmits over coaxial cable. The serializer also supports the bidirectional control data communication between the camera and host controller. The on-board electronics of NileCAM81_CUOAGX serializer recovers the power supplied from the host controller without contaminating the data transfers.

NileCAM81_CUOAGX board has been designed and developed by e-con Systems for the NVIDIA® Jetson AGX Xavier™ and NVIDIA® Jetson AGX Orin™ development kits. NileCAM81_CUOAGX is an 8 MP GMSL camera that contains a 1/1.7" AR0821 CMOS image sensor from onsemi™ along with on-board ISP. The coaxial cable is connected to serializer board through a rugged FAKRA connector. NileCAM81_CUOAGX can be directly interfaced with Jetson AGX Xavier™/Orin™ development kit through J509 connector.

The NVIDIA® Jetson AGX Xavier™/Orin™ development kit is a full-featured development platform for visual computing. It is ideal for applications requiring high computational performance in a low power envelope.

e-con Systems also provides ecam_tk1_guvcview sample application that demonstrates the features of this camera. However, this camera can utilize any Video for Linux version 2 (V4L2) application.

The commands and output messages in this manual are represented by different colors as shown in below table.

Description

NileCAM81_CUOAGX can stream the resolutions and frame rates as shown in below table.

Table 1: Supported Resolutions and Frame Rates

Format	Resolution	Frame Rate (DAY HDR Mode)	Frame Rate (NIGHT HDR Mode)	Frame Rate (Linear Mode)
UYVY	HD	30 fps	60fps	60 fps
	FHD	30 fps	60fps	60 fps
	4K	16 fps	16fps	16 fps

Note:

- The frame rates listed in the above table can be achieved easily in manual exposure.
- In auto exposure, maximum frame rate could be achieved with maximum lighting.

The camera controls of NileCAM81_CUOAGX are as follows:

- Brightness
- Contrast
- Saturation
- Gamma
- Gain
- Horizontal Flip
- Vertical Flip
- White Balance (both manual and automatic modes)
- Sharpness
- Exposure
- ROI Window Size
- HDR
- Trigger
- Frame Sync
- Trigger Mode
- Denoise
- Exposure Compensation
- Powerline Frequency Selection
- Frame Rate Control
- Special Effects

The e-multicam.elf is a simple Gstreamer application for viewing video from the devices supported on the Jetson AGX Xavier™/Orin™ development kit. This e-multicam.elf application can stream maximum of six cameras when connected with Jetson AGX Xavier™/Orin development kit.

Using e-multicam.elf application, you can perform the following:

- Enumerate and list all video capture devices connected.
- Stream all available resolutions if different resolutions are supported by the device.
- Display the average frame rate.
- Change controls for all available cameras.
- Capture images in raw, bmp and jpeg formats.
- Record video in H264 and UYVY format.

e-con Systems provides prebuilt binary of the application along with source code. Please refer to the *NileCAM81_CUOAGX_Release_Notes.pdf* for the compatible Linux distribution version (L4T version).

Identifying the Deliverables

This section describes about identifying the deliverables.

The release package contains the application source code. The commands and output messages in this manual are represented by different colors as shown in below table.

Table 2: Notation of Colors

Color	Notation
Blue	Commands running in development PC
Red	Output message in development PC
Green	Commands running in development board

The steps for identifying the deliverables are as follows:

The steps for identifying the deliverables are as follows:

1. Copy the release package tar file to the home directory of the board.
2. Run the following commands to extract the NileCAM81_CUOAGX release package.

```
tar -xaf e-CAM_YUV-GMSL-PRODUCTS_JETSON_<L4T_version>_<release_date>_<release_version>.tar.xz
cd e-CAM_YUV-GMSL-PRODUCTS_JETSON_<L4T_version>_<release_date>_<release_version>
```

Note: Do not right click to extract the package.

The source code for the e-multicam.elf application is present in the release package at the following location.

Application/e-multicam/Source/e-multicam.tar.gz

Note: Please refer to *Building and Installing e-multicam from Source* section to build application from the source and refer to *Launching the Application* section for using the application.

Building and Installing e-multicam from Source

This section describes about building and installing the e-multicam.elf application from the source.

The steps to build and install e-multicam.elf application from the source are as follows:

1. Run the following commands to install the dependency Gstreamer-1.0 libraries.

   ```
   $ sudo apt-get update
   $ sudo apt-get install libgstreamer-plugins-base1.0-dev libgstreamer1.0-dev libjpeg9-dev
   $ sudo apt-get install make
   ```

   Note: If the installation process stops with the **could not get lock /var/lib/dpkg/lock** message, run the following command to remove the file and proceed with the installation.

   ```
   $ sudo rm /var/lib/dpkg/lock
   ```

   If the installation process stops with **unmet dependencies** message, run the following command to solve the issue.

   ```
   $ sudo apt –fix-broken install
   ```

2. Run the following commands to enter the source directory.

   ```
   $ tar -xf e-multicam.tar.gz
   $ cd e-multicam/
   ```

3. Run the following make command to build the e-multicam application from the source.

   ```
   $ make
   ```

4. Run the following make install command to install the build application.

   ```
   $ sudo make install
   ```

   The application will be installed in **/usr/local/bin** location.

Launching the Application

This section describes about launching the e-multicam application.

The steps to launch the application are as follows:

1. Connect the NileCAM81_CUOAGX MIPI camera to the connector on the Jetson AGX Xavier™/Orin™ development kit.
2. Run the following command with your IP address of the Jetson AGX Xavier™/Orin™ development kit to create SSH session from a Linux PC.

   ```
   Ssh nvidia@<ip-address>
   ```

   The module drivers for NileCAM81_CUOAGX provided by e-con Systems will be loaded automatically during board boot.

3. Run the following command to check whether all the six cameras are initialized.

   ```
   $ sudo dmesg | grep -I "Detected ar0821 sensor"
   ```

   The output message appears as shown below.

   ```
   Detected ar0821 sensor
   Detected ar0821 sensor
   Detected ar0821 sensor
   Detected ar0821 sensor
   Detected ar0821 sensor
   Detected ar0821 sensor
   ```

   The output message indicates that all the six cameras are initialized properly.

4. Run the following command to check the presence of six video nodes.

   ```
   $ ls /dev/video*
   ```

   The output message appears as shown below.

   ```
   /dev/video0
   /dev/video1
   /dev/video2
   /dev/video3
   /dev/video4
   /dev/video5
   ```

5. Run the following commands to check and set the power mode to maximum for better performance.

   ```
   $ sudo nvpmodel -q
   $ sudo nvpmodel -m 0
   ```


6. Run the following Jetson clocks command to achieve maximum frame rate before launching the e-multicam application in Jetson AGX Xavier™/Orin™ development kit.

```
$ sudo jetson_clocks
```

7. Run the following command before launching the e-multicam application from host PC.

```
$ export DISPLAY=:0
```

8. Run the following command to launch e-multicam.elf application from terminal

```
$ e-multicam.elf
```

When application is launched, the menu control will appear in the host PC and video display window will appear on the display connected to the Jetson AGX Xavier™/Orin™ development kit.

To close the application, press **Ctrl+C** from the menu control window in host PC.

Application Features

This section describes about the e-multicam application features.

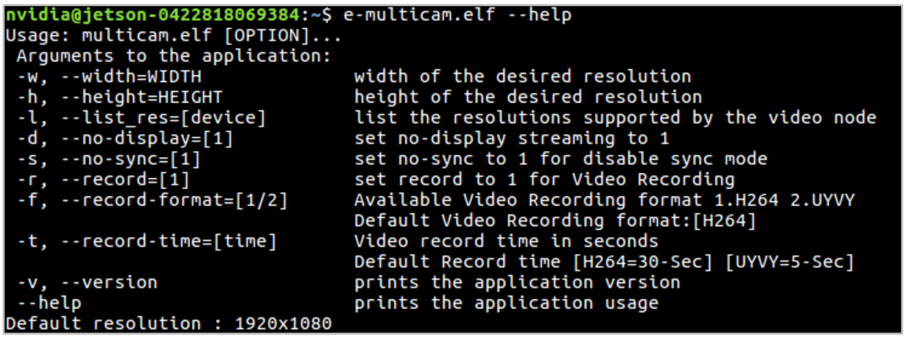
The features that are supported in the current version of e-multicam.elf application are as follows:

- Listing All Resolutions
- Streaming Required Resolution
- Displaying Image Controls
- Selecting Image Capture Format
- Video Recording Option

Run the following command to list the command line features.

```
$ e-multicam.elf -help
```

The features will appear as shown below.



```
nvidia@jetson-0422818069384:~$ e-multicam.elf --help
Usage: multicam.elf [OPTION]...
 Arguments to the application:
 -w, --width=WIDTH              width of the desired resolution
 -h, --height=HEIGHT            height of the desired resolution
 -l, --list_res=[device]        list the resolutions supported by the video node
 -d, --no-display=[1]           set no-display streaming to 1
 -s, --no-sync=[1]              set no-sync to 1 for disable sync mode
 -r, --record=[1]               set record to 1 for Video Recording
 -f, --record-format=[1/2]      Available Video Recording format 1.H264 2.UYVY
                                Default Video Recording format:[H264]
 -t, --record-time=[time]       Video record time in seconds
                                Default Record time [H264=30-Sec] [UYVY=5-Sec]
 -v, --version                  prints the application version
 --help                         prints the application usage
Default resolution : 1920x1080
```

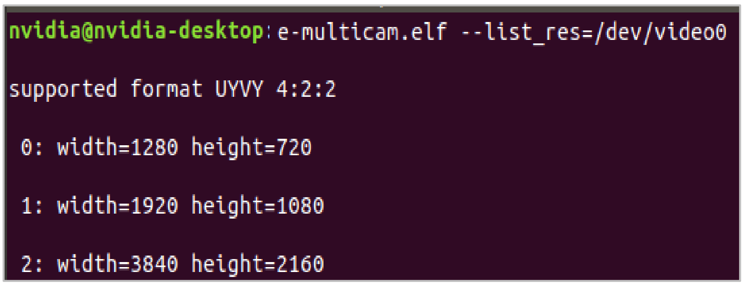
Figure 1: Command Line Features

Listing All Resolutions

Run the following command to view all the supported resolutions.

```
$ e-multicam.elf -list_res=/dev/video<n>
```

The supported resolutions will appear as shown below.



```
nvidia@nvidia-desktop: e-multicam.elf --list_res=/dev/video0
supported format UYVY 4:2:2
 0: width=1280 height=720
 1: width=1920 height=1080
 2: width=3840 height=2160
```

Figure 2: Supported Resolutions in UYVY Format

Or

Run the following command to view all the supported resolutions.

```
$ e-multicam.elf -l /dev/video<n>
```

The supported resolutions will appear as shown below.

```
nvidia@nvidia-desktop: e-multicam.elf -l /dev/video0
supported format UYVY 4:2:2
 0: width=1280 height=720
 1: width=1920 height=1080
 2: width=3840 height=2160
```

Figure 3: Supported Resolutions in UYVY Format

The supported resolutions in UYVY format are shown in below table.

Table 3: Supported Format and Resolution

Format	Resolution
UYVY	1280 x 720
	1920 x 1080
	3840 x 2160

Streaming Required Resolution

Run the following command to view the resolutions which are supported by the camera.

```
e-multicam.elf --width=<width> --height=<height>
```

The resolution will appear as shown below.

```
nvidia@nvidia-desktop:~$ e-multicam.elf --width=1280 --height=720

Arguments set:
        width  : 1280
        height : 720
```

Figure 4: Resolutions Supported by Camera

Or

Run the following command to view the resolutions which are supported by the camera.

```
e-multicam.elf -w 640 -h 480
```

The resolution will appear as shown below.

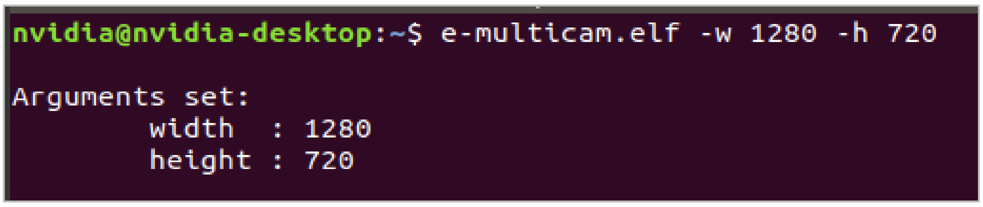


Figure 5: Resolutions Supported by Camera

Note: The default width is 1920 and default height is 1080. You must provide the supported resolution in the option as listed in *Table 3*.

Once the application is launched, the available cameras connected with the board are displayed as shown below.

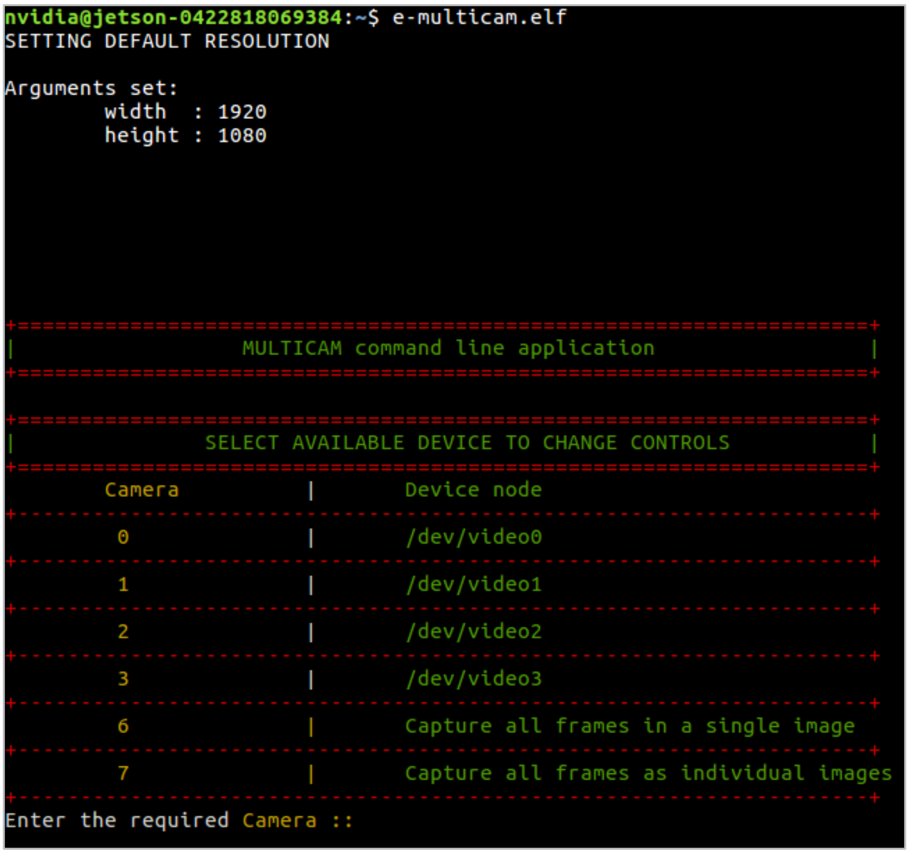


Figure 6: Displaying Cameras Connected with Board

You must select the camera for which the control value has to be changed.

No-display Option

If no-display option is set as 1, no frames will be displayed. By default, the application will display the frames.

Run the following command to set the no-display option.

```
e-multicam.elf --no-display 1
```

The no-display option will appear as shown below.


```
nvidia@jetson-0422818069384:~$ e-multicam.elf --no-display 1
SETTING DEFAULT RESOLUTION

Arguments set:
        width  : 1920
        height : 1080
```

Figure 7: No Display Option

Or

Run the following command to set the no-display option.

```
e-multicam.elf -d 1
```

The no-display option will appear as shown below.

```
nvidia@jetson-0422818069384:~$ e-multicam.elf -d 1
SETTING DEFAULT RESOLUTION

Arguments set:
        width  : 1920
        height : 1080
```

Figure 8: No Display Option

No-synchronous Option

If no-sync option is set as 1, the synchronous mode will be disabled. By default, the application will be in synchronous mode.

Run the following command to set the no-sync option.

```
e-multicam.elf --no-sync 1
```

or

```
e-multicam.elf -s 1
```

The no-sync option will appear as shown below.

```
nvidia@jetson-0422818069384:~$ e-multicam.elf --no-sync 1
SETTING DEFAULT RESOLUTION

Arguments set:
        width  : 1920
        height : 1080
```

or

```
nvidia@jetson-0422818069384:~$ e-multicam.elf -s 1
SETTING DEFAULT RESOLUTION

Arguments set:
        width  : 1920
        height : 1080
```

Displaying Image Controls

Enter the option to select the required camera, and you can view the supported controls of the selected camera as shown below.

```
+======================================================================+
                Image Controls Camera 0
+======================================================================+
Control | DESCRIPTION
+----------------------------------------------------------------------+
1       | Brightness
2       | Contrast
3       | Saturation
4       | Gamma
5       | Gain
6       | Horizontal Flip
7       | Vertical Flip
8       | White Balance Temperature
9       | Sharpness
10      | Exposure Time, Absolute
11      | ROI Window Size
12      | ROI Exposure
13      | HDR
14      | Frame Sync
15      | Denoise
16      | Exposure Compensation
17      | Powerline Frequency
18      | Frame Rate Control
19      | Special Effect
20      | TRIGGER
21      | Trigger_mode
22      | Display average frame rate
23      | Exit from Features Menu
+----------------------------------------------------------------------+
Enter the Control:
```

Figure 9: Displaying Supported Controls of Selected Camera

You can change the required control from the menu.

The image controls are as follows:

- Brightness
- Contrast
- Saturation
- Gamma
- Gain
- Horizontal Flip
- Vertical Flip
- White Balance
- Sharpness
- Exposure
- ROI Window Size
- HDR
- Trigger
- Frame Sync

- [Trigger Mode]
- [Denoise]
- [Exposure Compensation]
- [Powerline Frequency]
- [Frame rate control]
- [Special Effects]
- [Display Average Frame Rate]

The values of NileCAM81_CUOAGX controls are shown in below table.

Table 4: Values of NileCAM81_CUOAGX Controls

Controls	Minimum Value	Maximum Value	Default Value	Manual Control	Auto Control
Brightness	-15	15	0	YES	NO
Contrast	0	10	5	YES	NO
Saturation	0	60	16	YES	NO
White Balance	1000	10000	4500	YES	YES
Gamma	40	500	220	YES	NO
Gain	1	100	1	YES	NO
Horizontal Flip	0	1	0	YES	NO
Vertical Flip	0	1	0	YES	NO
Sharpness	0	7	2	YES	NO
Exposure	1 (100 µs)	10000 (1 second)	312 (31.2 ms)	YES	YES
ROI Window Size	8	64	8	YES	NO
HDR	Day HDR	Linear	Day HDR	YES	NO
Trigger	0 (INTERNAL TRIGGER)	1 (EXTERNAL TRIGGER)	0 (INTERNAL TRIGGER	YES	NO
Frame Sync Mode	0 (SYNC 15 HZ)	2 (SYNC 60 HZ)	1 (SYNC 30 HZ)	YES	NO
Trigger Mode	0 (DISABLED MODE)	2 (TRIGGER ONLY MODE)	1 (FREE RUNNING MODE)	YES	NO
Denoise	0	15	8	YES	NO
Exposure Compensation	8000	1000000	33333	YES	NO
Powerline Frequency	Auto	60HZ	Auto	YES	NO
Frame Rate Control	0	60	30	YES	No
Special Effects	Normal	Sketch	Normal	YES	No

Brightness

Enter the option 1 to view the Brightness Menu as shown below.

```
+==================================================================+
Brightness Menu -- Range -15 to 15, Step Size = 1, Default Value = 0
+==================================================================+
Value    | DESCRIPTION
+------------------------------------------------------------------+
 1       | Choose Brightness level
 2       | Exit from  Brightness menu
 Note:
Current Brightness Value is 0
+==================================================================+

  Please choose the Value :
```

Figure 10: Brightness Menu

The brightness values can be changed from a minimum value of -15 to 15, and the exact changes will be reflected immediately in the preview. This brightness control increases the low light performance of NileCAM81_CUOAGX. The default value is 0.

Contrast

Enter the option 2 to view the Contrast Menu as shown below.

```
Contrast Menu -- Range 0 to 10, Step Size = 1, Default Value = 5
+==================================================================+
Value    | DESCRIPTION
+------------------------------------------------------------------+
 1       | Choose Contrast level
 2       | Exit from  Contrast menu
 Note:
Current Contrast Value is 5
+==================================================================+

  Please choose the Value :
```

Figure 11: Contrast Menu

The contrast values can be changed from a minimum value of 0 to 10, and the exact changes will be reflected immediately in the preview. Increasing the contrast value increases the luminance of NileCAM81_CUOAGX. The default value is 5.

Saturation

Enter the option 3 to view the Saturation Menu as shown below.

```
+==================================================================+
Saturation Menu -- Range 0 to 60, Step Size = 1, Default Value = 16
+==================================================================+
Value    | DESCRIPTION
+------------------------------------------------------------------+
 1       | Choose Saturation level
 2       | Exit from  Saturation menu
 Note:
Current Saturation Value is 16
+==================================================================+

  Please choose the Value :
```

Figure 12: Saturation Menu

The saturation values can be changed from a minimum value of 0 to 60, and the exact changes will be reflected immediately in the preview. Increasing the saturation value increases the intensity of the color of NileCAM81_CUOAGX. The default value is 16.

Gamma

Enter the option 4 to view the Gamma Menu as shown below.

```
+==============================================================+
Gamma Menu -- Range 40 to 500, Step Size = 1, Default Value = 220
+==============================================================+
Value    | DESCRIPTION
+--------------------------------------------------------------+
 1       | Choose Gamma level
 2       | Exit from  Gamma menu
 Note:
Current Gamma Value is 220
+==============================================================+

  Please choose the Value :
```

Figure 13: Gamma Menu

The gamma values can be changed from a minimum value of 40 to 500, and the exact changes will be reflected immediately in the preview. The default value is 220.

Gain

Enter the option 5 to view the Gain Menu as shown below.

```
+==============================================================+
Gain Menu -- Range 1 to 100, Step Size = 1, Default Value = 1
+==============================================================+
Value    | DESCRIPTION
+--------------------------------------------------------------+
 1       | Choose Gain level
 2       | Exit from  Gain menu
 Note:
Current Gain Value is 1
+==============================================================+

  Please choose the Value :
```

Figure 14: Gain Menu

The gain values can be changed from a minimum of 1 to 100. The changes are updated in the preview only when the exposure control is set to manual mode. The default value is 1.

Horizontal Flip

Enter the option 6 to view the Horizontal Flip Menu as shown below.

```
+=================================================================+
Horizontal Flip Menu -- Range 0 to 1, Step Size = 1, Default Value = 0
+=================================================================+
Value   | DESCRIPTION
+-----------------------------------------------------------------+
 1      | Choose Horizontal Flip level
 2      | Exit from  Horizontal Flip menu
 Note:
Current Horizontal Flip Value is 0
+=================================================================+

  Please choose the Value :
1
Please enter the Horizontal Flip Value :
1
```

Figure 15: Horizontal Flip Menu

The preview from the sensor can be horizontally flipped by entering the value 1. By default, the value is 0.

Vertical Flip

Enter the option 7 to view the Vertical Flip Menu as shown below.

```
+=================================================================+
Vertical Flip Menu -- Range 0 to 1, Step Size = 1, Default Value = 0
+=================================================================+
Value   | DESCRIPTION
+-----------------------------------------------------------------+
 1      | Choose Vertical Flip level
 2      | Exit from  Vertical Flip menu
 Note:
Current Vertical Flip Value is 0
+=================================================================+

  Please choose the Value :
1
Please enter the Vertical Flip Value :
1
```

Figure 16: Vertical Flip Menu

The preview from the sensor can be vertically flipped by entering the value 1. By default, the value is 0.

White Balance

Enter the option 8 to view the White Balance Menu. You can enter the option 0 to set the manual white balance mode as shown below.

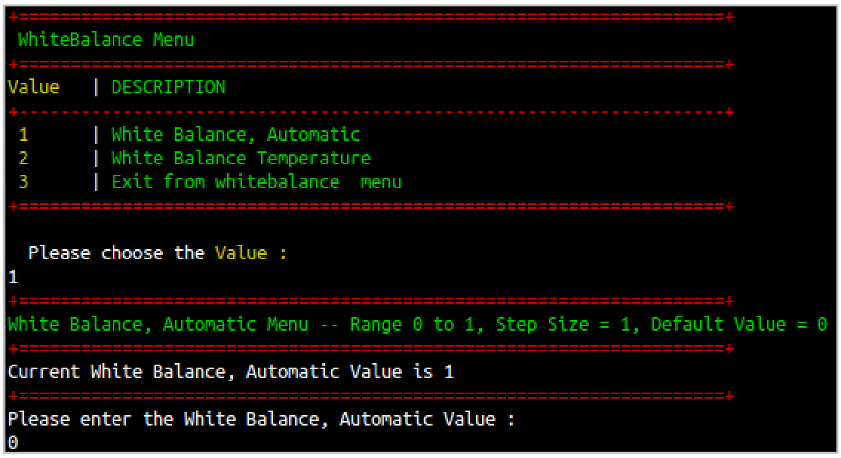


Figure 17: White Balance Menu

The manual white balance values can be changed from 1000 to 10000, and the exact changes will be reflected immediately in the preview. The default value is 4500. By default, the camera is in White Balance, Automatic mode as shown below.

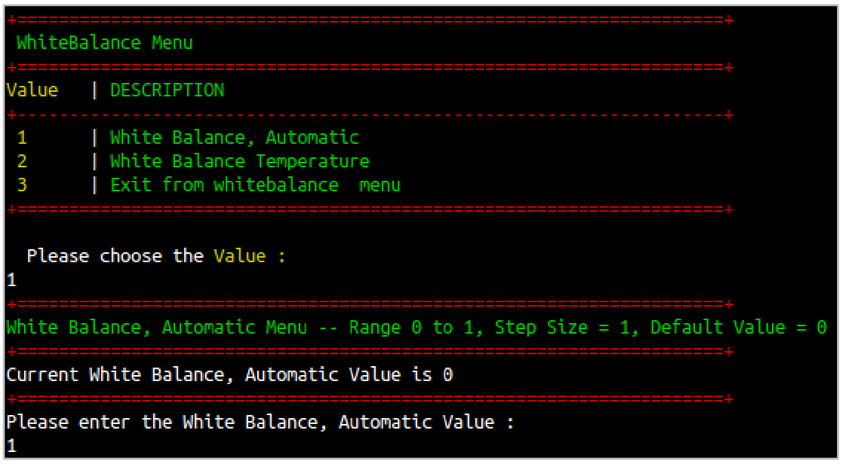


Figure 18: Camera in White Balance, Automatic Mode

Enter the option 2 to set the White Balance Temperature mode as shown below.

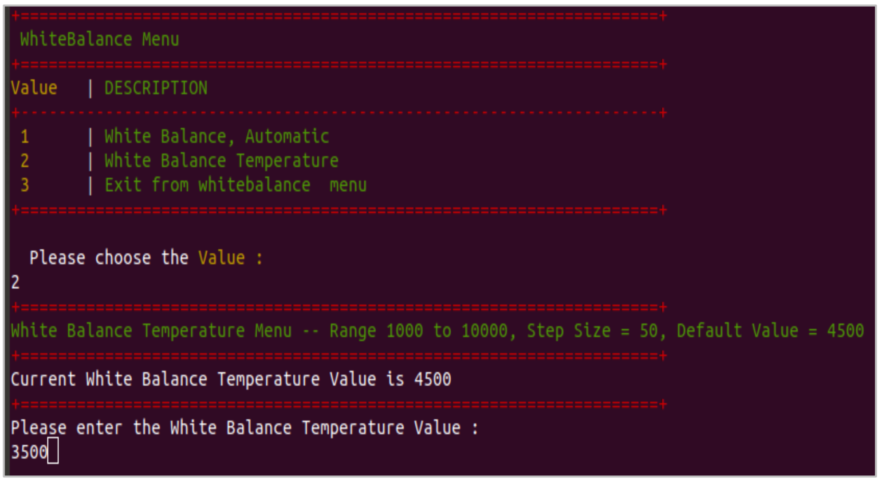


Figure 19: Camera in White Balance Temperature Mode

Sharpness

Enter the option 9 to view the Sharpness Menu as shown below.

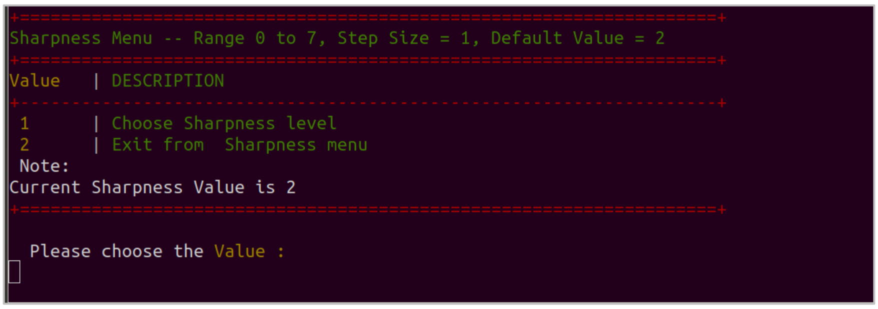


Figure 20: Sharpness Menu

The sharpness values can be changed from a minimum value of 0 to 7, and the exact changes will be reflected immediately in the preview. This sharpness control increases clarity of NileCAM81_CUOAGX. The default value is 2.

Exposure

NileCAM81_CUOAGX supports manual, auto and ROI based exposure mode. The default exposure mode is manual mode.

The modes of exposure control are shown in below table.

Table 5: Modes of Exposure Control

Values	Exposure Mode
0	Auto Exposure
1	Manual Exposure
2	ROI Based Exposure

In ROI based exposure mode, you can select the ROI and at that region, the exposure value will be applied to the entire frame. To select ROI based auto mode, you must enter the option 2.

You can enter the option 1 to set the manual exposure mode as shown below.

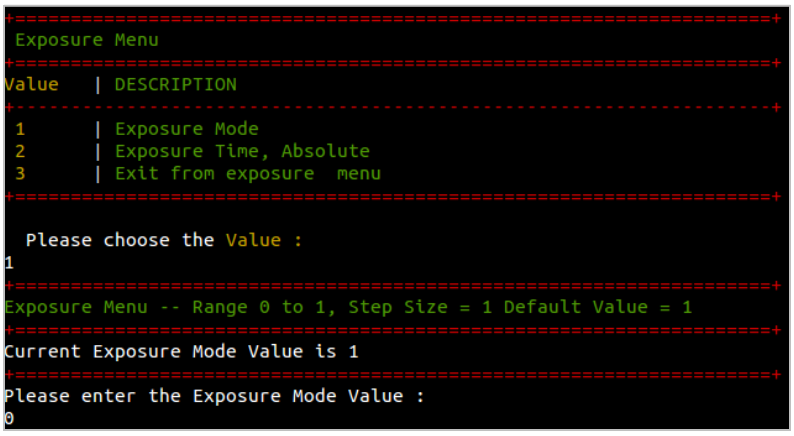


Figure 21: Camera in Manual Exposure Mode

The exposure value could be manually changed by entering the exposure value, and NileCAM81_CUOAGX supports exposure values ranging from 100 µs to 1 second represented from 1 to 10000 as shown below.

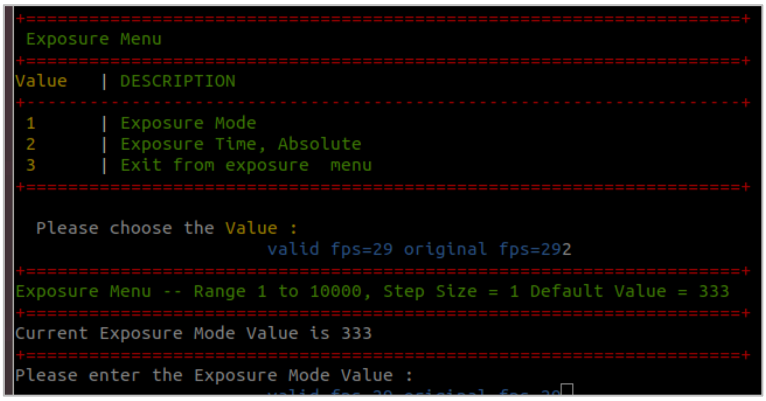


Figure 22: Exposure Values

The exposure values are configured inside the CMOS image sensor based on the sensor configuration and clock configuration details.

The exposure value applied in the sensor in ms is 1/10 of the set value as shown in below table.

Table 6: Exposure Value-Exposure Time Mapping

Exposure Value	Exposure Time
1	0.1 ms
2	0.2 ms
3	0.3 ms
..	..
10	1 ms
11	1.1 ms
12	1.2 ms
..	..

100	10 ms
..	..
1000	100 ms
..	..
7500	750 ms
10000	1 second

Note: When the exposure time is more than the time period of camera frame, the frame rate will drop.

To obtain a good low light performance it is essential to change the exposure according to the change in lighting conditions. To support this feature, NileCAM81_CUOAGX has an auto exposure mode, by which the exposure of the camera will be changed according to the lighting conditions giving the best low light performance.

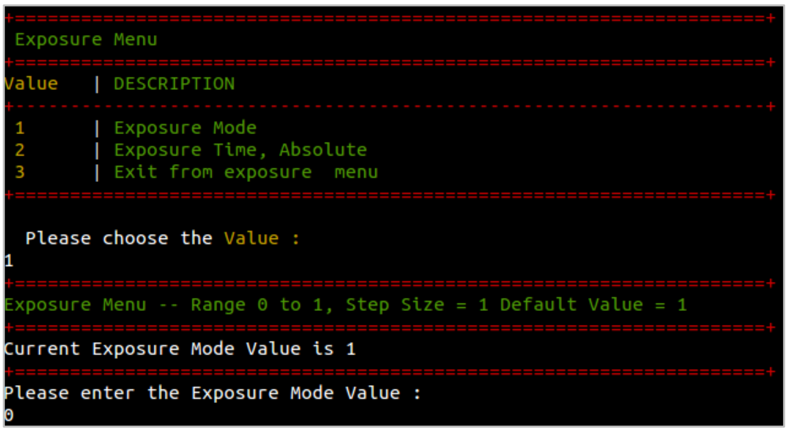


Figure 23: Auto Exposure Mode

ROI Window Size

You can change the ROI window size, by entering a suitable value as input. The range is from 8 to 64 with the step size of 8. The default value is 8.

Enter the option 11 to view the ROI Window size Menu as shown below.

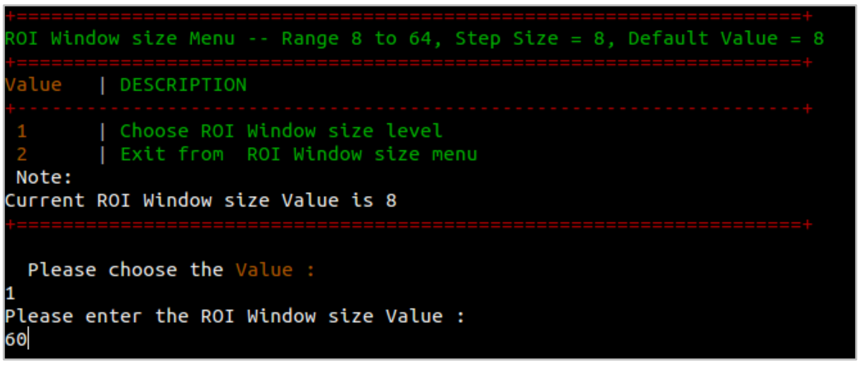


Figure 24: ROI Window Size Menu

For window size 64, the entire frame will be the ROI. If this region exceeds or falls below the frame boundary, the ROI will be clipped automatically.

For 1280 x 720 frame size, the ROI based on the window size is shown in below table.

Table 7: Window Size - ROI Region

Window Size	ROI Region
8	1/8 (160 x 90)
16	2/8 (320 x 180)
24	3/8 (480 x 270)
32	4/8 (640 x 360)
40	5/8 (800 x 450)
48	6/8 (960 x 540)
56	7/8 (1120 x 630)
64	1 (1280 x 720)

HDR

In HDR, there are three available modes as follows:

- **Day HDR**: When the sensor is operated at HDR mode, the dynamic range will be greater than linear mode, recommended to use in a scene with good indoor light or sunlight. To enable this mode, you select Day HDR in HDR control by enter value 0.
- **Night HDR:** When the sensor is operated at HDR mode, the dynamic range will be greater than linear mode, recommended to use in a scene with low light. To enable this mode, you select Night HDR in HDR control by enter value 1.
- **Linear:** When the sensor is operated at linear mode, the dynamic range of the device will be in normal range. To enable this mode, you select Linear in HDR control by enter value 2.

Enter the option 13 to view the HDR Menu as shown below.

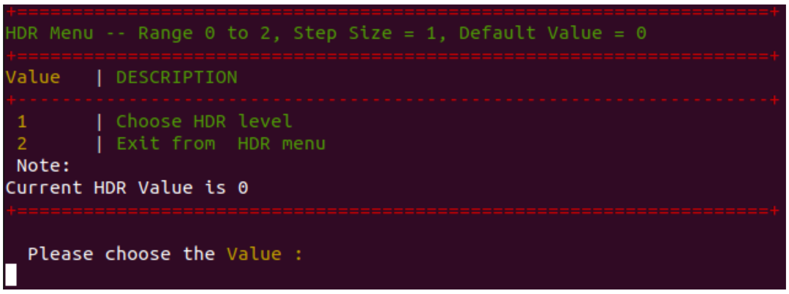
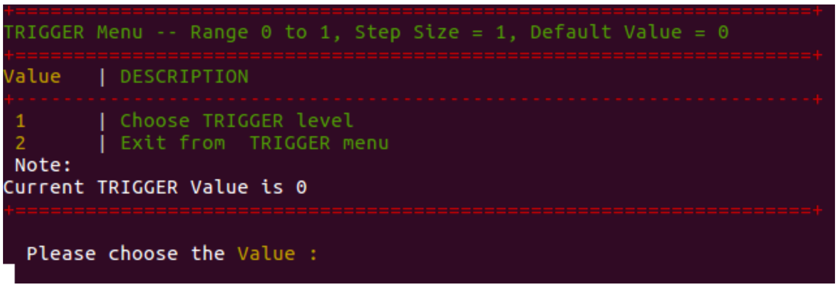


Figure 25: HDR Menu

Trigger

Enter the option 14 to view the TRIGGER Menu as shown below.



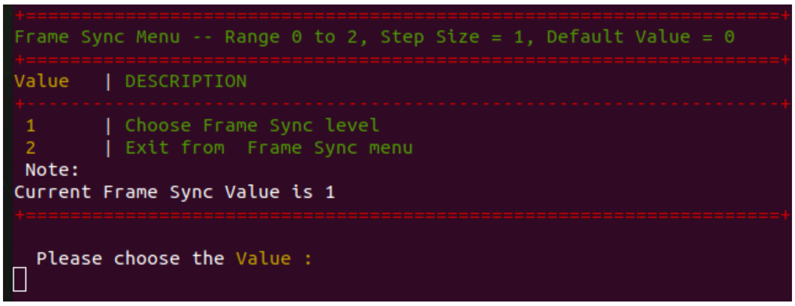
```
+==================================================================+
TRIGGER Menu -- Range 0 to 1, Step Size = 1, Default Value = 0
+==================================================================+
Value    | DESCRIPTION
+------------------------------------------------------------------+
 1       | Choose TRIGGER level
 2       | Exit from  TRIGGER menu
 Note:
Current TRIGGER Value is 0
+==================================================================+

  Please choose the Value :
```

Figure 26: Trigger Menu

The TRIGGER Menu provide the option to switch between internal and external trigger. By default, TRIGGER Menu is in internal trigger. To toggle TRIGGER Menu, frame sync mode should be enable.

Frame Sync Mode

Enter the option 15 to view the Frame Sync Mode Menu as shown below.



```
+==================================================================+
Frame Sync Menu -- Range 0 to 2, Step Size = 1, Default Value = 0
+==================================================================+
Value    | DESCRIPTION
+------------------------------------------------------------------+
 1       | Choose Frame Sync level
 2       | Exit from  Frame Sync menu
 Note:
Current Frame Sync Value is 1
+==================================================================+

  Please choose the Value :
```

Figure 27: Frame Sync Menu

You can choose the frame sync control by setting values as 0 (enable, synchronous mode at 15fps), 1 (enable, synchronous mode at 30fps), 2 (enable, synchronous mode at 60fps). The default value is 1.

The frame sync control value and its modes are listed in below table.

Table 8: Frame Sync Control Value

Control value	Frame Sync Mode
0	Synchronous mode with 15 fps
1	Synchronous mode with 30 fps
2	Synchronous mode with 60 fps

Note:

- By default, the frame sync value is 0, which is then changed to 1 during launch of application.
- All Camera must be in a same HDR before switching to frame sync mode.
- If the frame sync value is changed for one camera, it will be automatically applied to other connected cameras. Therefore, there is no need to change the frame sync value for all the cameras.
- The readout times of the frames may vary if each camera is set to a different exposure time.
- For 4k Synchronous mode we must enable the external trigger.

Trigger Mode

In Trigger mode, there are three available modes as follows:

- **Disabled mode**: When the sensor is operated at Disabled mode, the camera ignores the trigger and output frames continuously. To enable this mode, you select Disabled mode in Trigger mode control by enter value 0.
- **Free running mode:** When the sensor is operated at Free running mode, the camera frames are sync to trigger only when trigger signal is detected, otherwise runs freely. To enable this mode, you select Free Running Mode in Trigger mode control by enter value 1.
- **Trigger only mode:** When the sensor is operated at Trigger Only mode, the camera outputs the frame only when trigger signal is detected. To enable this mode, you select Trigger Only mode in Trigger mode control by enter value 2.

Enter the option 16 to view the Trigger mode Menu as shown below.

```
+==================================================================+
Trigger_mode Menu -- Range 0 to 2, Step Size = 1, Default Value = 1
+==================================================================+
Value    | DESCRIPTION
+------------------------------------------------------------------+
 1       | Choose Trigger_mode level
 2       | Exit from  Trigger_mode menu
 Note:
Current Trigger_mode Value is 0
+==================================================================+

  Please choose the Value :
```

Figure 28: Trigger Menu

Denoise

The denoise control is used to remove pixel noise from the preview. This control value can be changed from a minimum value of 0 to 15 by moving the slider. The default value is 8.

Enter the option 17 to view the Denoise Menu as shown below.

```
+==================================================================+
Denoise Menu -- Range 0 to 15, Step Size = 1, Default Value = 8
+==================================================================+
Value    | DESCRIPTION
+------------------------------------------------------------------+
 1       | Choose Denoise level
 2       | Exit from  Denoise menu
 Note:
Current Denoise Value is 8
+==================================================================+

  Please choose the Value :
```

Figure 29: Denoise Menu

Exposure Compensation

The exposure compensation control adjusts the upper limit of auto exposure. The range is from 8000µs to 1000000µs. The default value is 33333 µs.

Enter the option 18 to view the Exposure compensation menu as shown below.

```
+==================================================================+
Exposure Compensation Menu -- Range 8000 to 1000000, Step Size = 1, Default Value = 33333
+==================================================================+
Value    | DESCRIPTION
+------------------------------------------------------------------+
 1       | Choose Exposure Compensation level
 2       | Exit from  Exposure Compensation menu
 Note:
Current Exposure Compensation Value is 33333
+==================================================================+

  Please choose the Value :
```

Figure 30: Exposure Compensation Menu

Powerline Frequency

The powerline frequency values can be selected between 50 Hz and 60 Hz. The default mode is auto.

Enter the option 19 to view the Powerline Frequency Menu as shown below.

```
+==================================================================+
Powerline Frequency Menu -- Range 0 to 2, Step Size = 1, Default Value = 0
+==================================================================+
Value    | DESCRIPTION
+------------------------------------------------------------------+
 1       | Choose Powerline Frequency level
 2       | Exit from  Powerline Frequency menu
 Note:
Current Powerline Frequency Value is 0
+==================================================================+

  Please choose the Value :
```

Figure 31: Powerline Frequency Menu

Frame Rate control

You can change the frame rate using the Frame Rate Control. The default frame rate will display depends upon HDR.

Note : This control will be effect only if frame sync option is disabled. By default, the value of frame sync control is 2, change the frame sync control to 0 for disable it, then change the frame rate control. Otherwise, there will be no effect for this control.

Enter the option 20 to view the Frame Rate control Menu as shown below.

```
+====================================================================+
Frame Rate Control Menu -- Range 3 to 60, Step Size = 1, Default Value = 30
+====================================================================+
Value    | DESCRIPTION
+--------------------------------------------------------------------+
 1       | Choose Frame Rate Control level
 2       | Exit from  Frame Rate Control menu
 Note:
Current Frame Rate Control Value is 30
+====================================================================+

  Please choose the Value :
```

Figure 32: Frame Rate control Menu

Special Effects

The special effects enable four other processed image preview options in NileCAM81_CUOAGX.

Enter the option 21 to view the Special Effects Menu as shown below.

```
18
+====================================================================+
Special Effect Menu -- Range 0 to 4, Step Size = 1, Default Value = 0
+====================================================================+
Value    | DESCRIPTION
+--------------------------------------------------------------------+
 1       | Choose Special Effect level
 2       | Exit from  Special Effect menu
 Note:
Current Special Effect Value is 0
+====================================================================+

  Please choose the Value :
```

Figure 33: Frame Rate control Menu

Display Average Frame Rate

Enter the option 22 to view the Display Average Frame Rate menu. You can enable or disable the average fps indication using this control. On enabling the average fps indication, a debug message appears as shown below.

```
+=====================================================================+
 Show Frame rate debug message
+=====================================================================+
Value   | DESCRIPTION
+---------------------------------------------------------------------+
 0      | Disable
 1      | Enable
 2      | Exit from Show Frame rate menu
+---------------------------------------------------------------------+
 Note:
 Current Value is 0
+=====================================================================+
 Enter the Value :
1
+=====================================================================+
 Show Frame rate debug message
+=====================================================================+
Value   | DESCRIPTION
+---------------------------------------------------------------------+
 0      | Disable
 1      | Enable
 2      | Exit from Show Frame rate menu
+---------------------------------------------------------------------+
 Note:
 Current Value is 1
+=====================================================================+
 Enter the Value :
 valid fps=29 original fps=29
```

Figure 34: Debug Message

Selecting Image Capture Format

You can capture images by entering the option 6 or 7 available along with the listing of camera devices. After entering the option, the format in which the image must be saved will be displayed as shown below.

```
Enter the required Camera ::

7
+======================================================================+

Individual Image capture selected
+======================================================================+
                Image formats Available
+======================================================================+
         1              |          Raw Image Format
+======================================================================+
         2              |          BMP Image Format
+======================================================================+
         3              |          JPEG Image Format
+======================================================================+

Enter the required format:
3
File saved: /home/nvidia/image-14-13-29_0.jpg

Image captured..
```

Figure 35: Displaying Image Format

The formats available to save the image are as follows.

- Raw format
- BMP format
- JPEG format

If the image capture is successful, the image file name along with the path will be displayed and the **Image capture done** message will be displayed.

Capturing Frame in Single Image File

You can enter the option 6 to capture all the Six frames in a single image file. This menu has the option to choose the image format to which the frames get stored. The resolution of the image in which all frames are captured together is based on the resolution in which the application is streaming as shown in below table.

Table 8: Camera Connection Based Image Resolution

Number of Cameras Connected	Resolution
1	(current_frame_width) x (current_frame_height)
2	(current_frame_width) x (current_frame_height x 2)
3	(current_frame_width) x (current_frame_height x 3)
4	(current_frame_width x 2) x (current_frame_height x 2)
5	(current_frame_width x 2) x (current_frame_height x 3)
6	(current_frame_width x 2) x (current_frame_height x 3)

The captured images will get stored in the home directory with **image-<timestamp>_<connected_camera>.<format>** file name.

Note: <timestamp> represented in hh-mm-ss format denotes the system time (hour-minute-second) during the image capture and <connected_camera> represented the total number of cameras connected with the Jetson Xavier™/AGX Orin™ board.

Capturing Frame in Individual Image File

You can enter the option 7 to capture each frame in individual image files. The resolution of the individual images captured are same as the resolution selected for streaming.

The captured images will get stored in the home directory with **image-<timestamp>_<camera_number>.<format>** file name.

Note: <timestamp> represented in hh-mm-ss format denotes the system time (hour-minute-second) during the image capture.

For example,

Raw Image: <HOME_PATH>/image-14-13-29_0.yuv

BMP Image: <HOME_PATH>/ image-16-46-21_2.bmp

JPEG Image: <HOME_PATH>/image-17-03-12_1.jpg

Note: Image capture option is available only when you exit from feature menu in image controls, by entering the option 20 in image controls.

Video Recording Option

The e-multicam.elf application supports the video recording in H264 and UYVY format. When streaming starts with video record option, the camera preview will not be available.

Run the following command to record video in default configuration.

```
$ e-multicam.elf -r 1
```

The progress of video recording will appear as shown below.

```
nvidia@jetson-0422818069384:~$ e-multicam.elf -r 1
################# Video Recording in Progress ... ####################
SETTING DEFAULT RESOLUTION

Arguments set:
        width  : 1920
        height : 1080

Framerate set to : 30 at NvxVideoEncoderSetParameterNvMMLiteOpen : Block : BlockType = 4
===== NVMEDIA: NVENC =====
Framerate set to : 30 at NvxVideoEncoderSetParameterNvMMLiteOpen : Block : BlockType = 4
===== NVMEDIA: NVENC =====
NvMMLiteBlockCreate : Block : BlockType = 4
NvMMLiteBlockCreate : Block : BlockType = 4
H264: Profile = 66, Level = 40
H264: Profile = 66, Level = 40
Framerate set to : 30 at NvxVideoEncoderSetParameterNvMMLiteOpen : Block : BlockType = 4
===== NVMEDIA: NVENC =====
NvMMLiteBlockCreate : Block : BlockType = 4
H264: Profile = 66, Level = 40
```

Figure 36: Progress of Video Recording

Or

Run the following command to record video in default configuration.

```
$ e-multicam.elf --record 1
```

The progress of video recording will appear as shown below.

```
nvidia@jetson-0422818069384:~/Videos$ e-multicam.elf --record 1
################# Video Recording in Progress ... ####################
SETTING DEFAULT RESOLUTION

Arguments set:
        width  : 1920
        height : 1080

Framerate set to : 30 at NvxVideoEncoderSetParameterNvMMLiteOpen : Block : BlockType = 4
===== NVMEDIA: NVENC =====
NvMMLiteBlockCreate : Block : BlockType = 4
Framerate set to : 30 at NvxVideoEncoderSetParameterNvMMLiteOpen : Block : BlockType = 4
===== NVMEDIA: NVENC =====
NvMMLiteBlockCreate : Block : BlockType = 4
H264: Profile = 66, Level = 40
H264: Profile = 66, Level = 40
Framerate set to : 30 at NvxVideoEncoderSetParameterNvMMLiteOpen : Block : BlockType = 4
===== NVMEDIA: NVENC =====
NvMMLiteBlockCreate : Block : BlockType = 4
H264: Profile = 66, Level = 40
```

Figure 37: Progress of Video Recording

The default video recording is H264 encoded format for 30 seconds.

The recorded video file will be available in <HOME_PATH>**/Videos** directories with **video-<timestamp>_<camera_number>.<format>** file name.

For example,

video-7-14-45_0.mkv

video-7-14-46_1.mkv

video-7-14-46_2.mkv

Video Recording in H264 Format

Run the following command to record the video in H264 format.

```
$ e-multicam.elf -r 1 -f 1
```

The progress of H264 video recording will appear as shown below.

```
nvidia@jetson-0422818069384:~$ e-multicam.elf -r 1 -f 1
################# Video Recording in Progress ... ####################
SETTING DEFAULT RESOLUTION

Arguments set:
        width  : 1920
        height : 1080

Framerate set to : 30 at NvxVideoEncoderSetParameterNvMMLiteOpen : Block : BlockType = 4
===== NVMEDIA: NVENC =====
NvMMLiteBlockCreate : Block : BlockType = 4
Framerate set to : 30 at NvxVideoEncoderSetParameterNvMMLiteOpen : Block : BlockType = 4
===== NVMEDIA: NVENC =====
NvMMLiteBlockCreate : Block : BlockType = 4
H264: Profile = 66, Level = 40
H264: Profile = 66, Level = 40
Framerate set to : 30 at NvxVideoEncoderSetParameterNvMMLiteOpen : Block : BlockType = 4
===== NVMEDIA: NVENC =====
NvMMLiteBlockCreate : Block : BlockType = 4
H264: Profile = 66, Level = 40
```

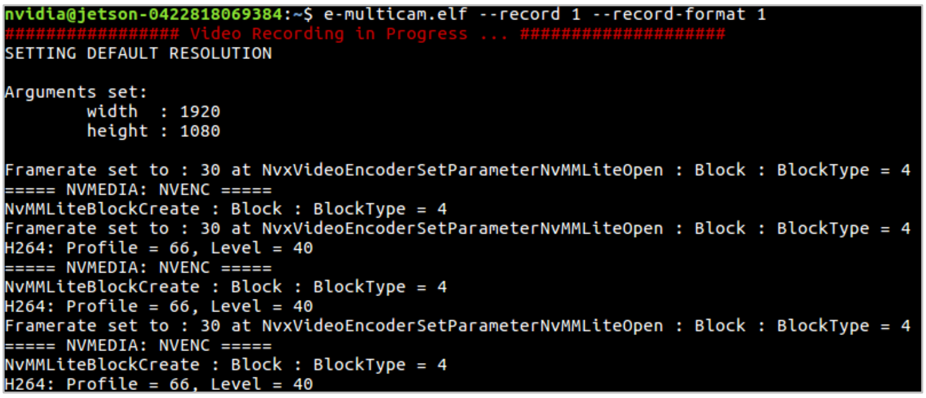
Figure 38: Progress of H264 Video Recording

Or

Run the following command to record the video in H264 format.

```
$ e-multicam.elf --record 1 --record-format 1
```

The progress of H264 video recording will appear as shown below.



```
nvidia@jetson-0422818069384:~$ e-multicam.elf --record 1 --record-format 1
################# Video Recording in Progress ... ####################
SETTING DEFAULT RESOLUTION

Arguments set:
        width  : 1920
        height : 1080

Framerate set to : 30 at NvxVideoEncoderSetParameterNvMMLiteOpen : Block : BlockType = 4
===== NVMEDIA: NVENC =====
NvMMLiteBlockCreate : Block : BlockType = 4
Framerate set to : 30 at NvxVideoEncoderSetParameterNvMMLiteOpen : Block : BlockType = 4
H264: Profile = 66, Level = 40
===== NVMEDIA: NVENC =====
NvMMLiteBlockCreate : Block : BlockType = 4
H264: Profile = 66, Level = 40
Framerate set to : 30 at NvxVideoEncoderSetParameterNvMMLiteOpen : Block : BlockType = 4
===== NVMEDIA: NVENC =====
NvMMLiteBlockCreate : Block : BlockType = 4
H264: Profile = 66, Level = 40
```

Figure 39: Progress of H264 Video Recording

Note: The default time for H264 video recording is 30 seconds.

Run the following command to change the default time.

```
$ e-multicam.elf -r 1 -f 1 -t <time>
```

The progress of H264 video recording with record time will appear as shown below.

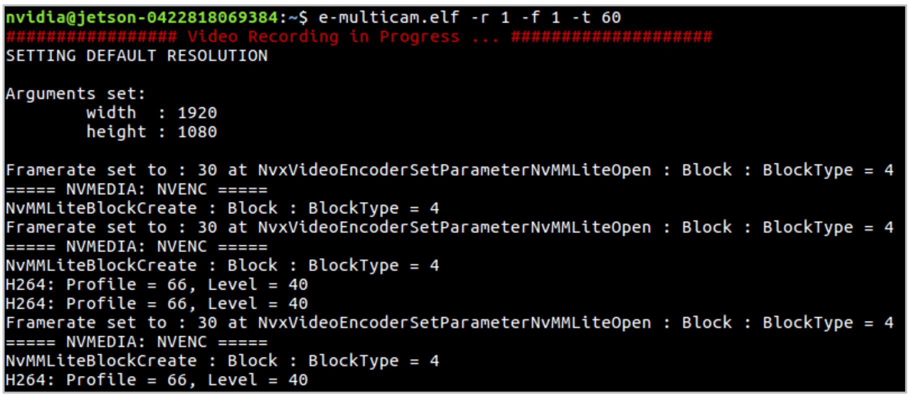


Figure 40: Progress of H264 Video Recording

Or

Run the following command to change the default time.

```
$ e-multicam.elf --record 1 --record-format 1 --record-
time <time>
```

Note: <time> must be in seconds.

The progress of H264 video recording with record time will appear as shown below.

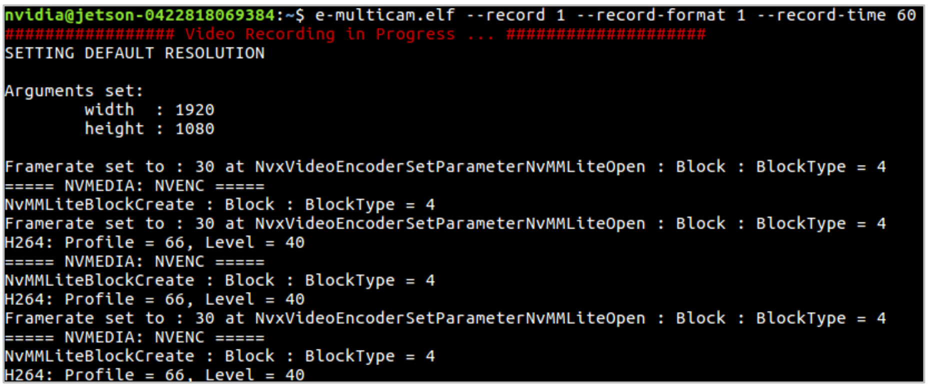


Figure 41: Progress of H264 Video Recording with Record Time

Note: Recording more than 1 min will reduce the frame rate.

Playback H264 Recorded Video

The recorded video can be played using Gstreamer command.

Run the following commands to playback the H264 recorded video.

```
$ export DISPLAY=:0
```



```
$ gst-launch-1.0 filesrc
location=<HOME_PATH>/Videos/video-<hr>-<min>-
<sec>_<cam_num>.mkv ! ! matroskademux ! h264parse !
nvv4l2decoder ! nv3dsink
```

Video Recording in UYVY Format

Run the following command to record video in UYVY format.

```
$ e-multicam.elf -r 1 -f 2
```

The progress of UYVY video recording will appear as shown below.

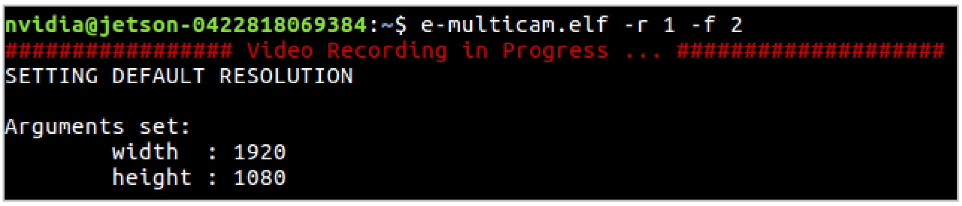


Figure 42: Progress of UYVY Video Recording

Or

Run the following command to record video in UYVY format.

```
$ e-multicam.elf --record 1 --record-format 2
```

The progress of UYVY video recording will appear as shown below.

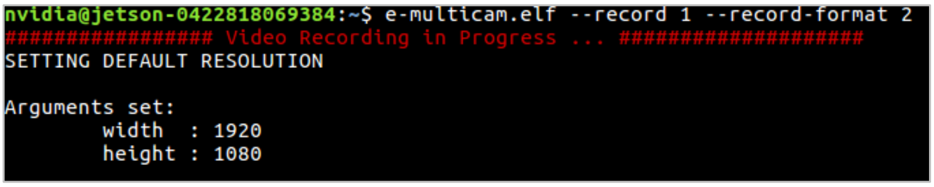


Figure 43: Progress of UYVY Video Recording

The default time for UYVY is 5 seconds.

Note: You must avoid UYVV recording for long time, because for 5 second recording it will take around 1.5 GB memory for 1080P resolution and this may occupy the entire eMMC. This recording option is provided for debugging purpose.

Playback UYVY Recorded Video

Run the following commands to playback the UYVY recorded video.

```
$ sudo apt-get update
$ sudo apt-get install mplayer
$ export DISPLAY=:0
$ mplayer <HOME_PATH>/Videos/video-<hr>-<min>-<sec>-cam.uyvy -demuxer rawvideo -rawvideo w=1920:h=1080:fps=30:format=uyvy
```

Note: The width and height vary based on recorded resolution.

Troubleshooting

In this section, you can view the list of commonly occurring issues and their troubleshooting steps.

The e-multicam.elf application is tested in HD+, FHD and 4K resolutions only. The frame rates may vary in other display resolutions because of resizing in application.

This is a known issue. Please write to techsupport@e-consystems.com to get immediate support on other issues.

I can view tearing issue in recorded 4K Video.

This is a known issue. Please write to techsupport@e-consystems.com to get immediate support on other issues.

I can view application termination which takes more time when more than one camera connected to Jetson AGX Xavier™/Orin™ board.

This is a known issue. Please write to techsupport@e-consystems.com to get immediate support on other issues.

1. **Why the overall frame rate of the application is lower than the mentioned frame rate?**

   The application overall frame rate depends on the exposure time set in all available camera modules and if they differ, the frame rate will vary accordingly.

   For example, if three cameras have exposure time as 30 ms and one camera is set to 100 ms, then the overall frame rate will be 10 fps.

   Note: Make sure to keep all the cameras to have the same exposure values to avoid this scenario.

2. **How can I get the updated package?**

   Please login to Developer Resources website and download the latest release package.

3. **Why I can't get 4K synchronous mode in default while launching the application?**

   4K Synchronous mode requires 15Hz, By default Jetson provides minimum of 30 Hz as an input trigger as we can only get 30fps and 60fps. If we need 4K 15fps in Synchronous mode we must enable the external trigger

   Please refer the below Document for external Trigger

   - *NileCAM81_CUOAGX External Trigger Setup Guide.*

What's Next?

After understanding the usage of e-multicam application, you can refer to the following documents to understand more about NileCAM81_CUOAGX.

- *NileCAM81_CUOAGX Release Notes*
- *NileCAM81_CUOAGX Linux App User Manual*

Glossary

4K: UHD or Industry name for 3840 x 2160 resolution.

API: Application Program Interface.

CMOS: Complementary Metal Oxide Semiconductor.

eMMC: embedded Multi-Media Card.

FHD: Full HD (Industry name for 1920 x 1080 resolution).

HD: High Definition (Industry name for 1280 x 720 resolution).

HDR: High Dynamic Range.

L4T: Linux for Tegra.

MIPI: Mobile Industry Processor Interface.

ROI: Region of Interest.

SSH: Secure Shell.

UHD: Ultra HD (Industry name for 3840 x 2160 resolution).

UYVY: YUV422 16-bit image format with UYVY ordering.

V4L2: Video for Linux version 2 is a collection of device drivers and API for supporting real-time video capture on Linux systems.

Support

Contact Us

If you need any support on NileCAM81_CUOAGX product, please contact us using the Live Chat option available on our website - https://www.e-consystems.com/

Creating a Ticket

If you need to create a ticket for any type of issue, please visit the ticketing page on our website - https://www.e-consystems.com/create-ticket.asp

RMA

To know about our Return Material Authorization (RMA) policy, please visit the RMA Policy page on our website - https://www.e-consystems.com/RMA-Policy.asp

General Product Warranty Terms

To know about our General Product Warranty Terms, please visit the General Warranty Terms page on our website - https://www.e-consystems.com/warranty.asp

Revision History

Rev	Date	Description	Author
1.0	02-Nov-2022	Initial draft	Camera Dev Team
1.1	23-Jan-2023	Updated Sensor format from 1/2 to 1/1.7	Camera Dev Team
1.2	30-Jan-2023	Recorded video playback command modified and minor changes	Camera Dev Team
1.3	10-Feb-2023	Added Sync Support	Camera Dev Team
1.4	22-Mar-2023	Updated Contrast Value	Camera Dev Team
1.5	24-May-2023	Updated Frame Rate	Camera Dev Team
1.6	07-Jun-2023	Added Trigger Mode Control	Camera Dev Team
1.7	08-Jan-2024	Renamed Cam mode to HDR and renamed the release package	Camera Dev Team